Learning Spatial Unit-Norm Latents with Riemannian Flow Matching

SRUL: Spherical Riemannian Unit-Norm Latents

Motivation

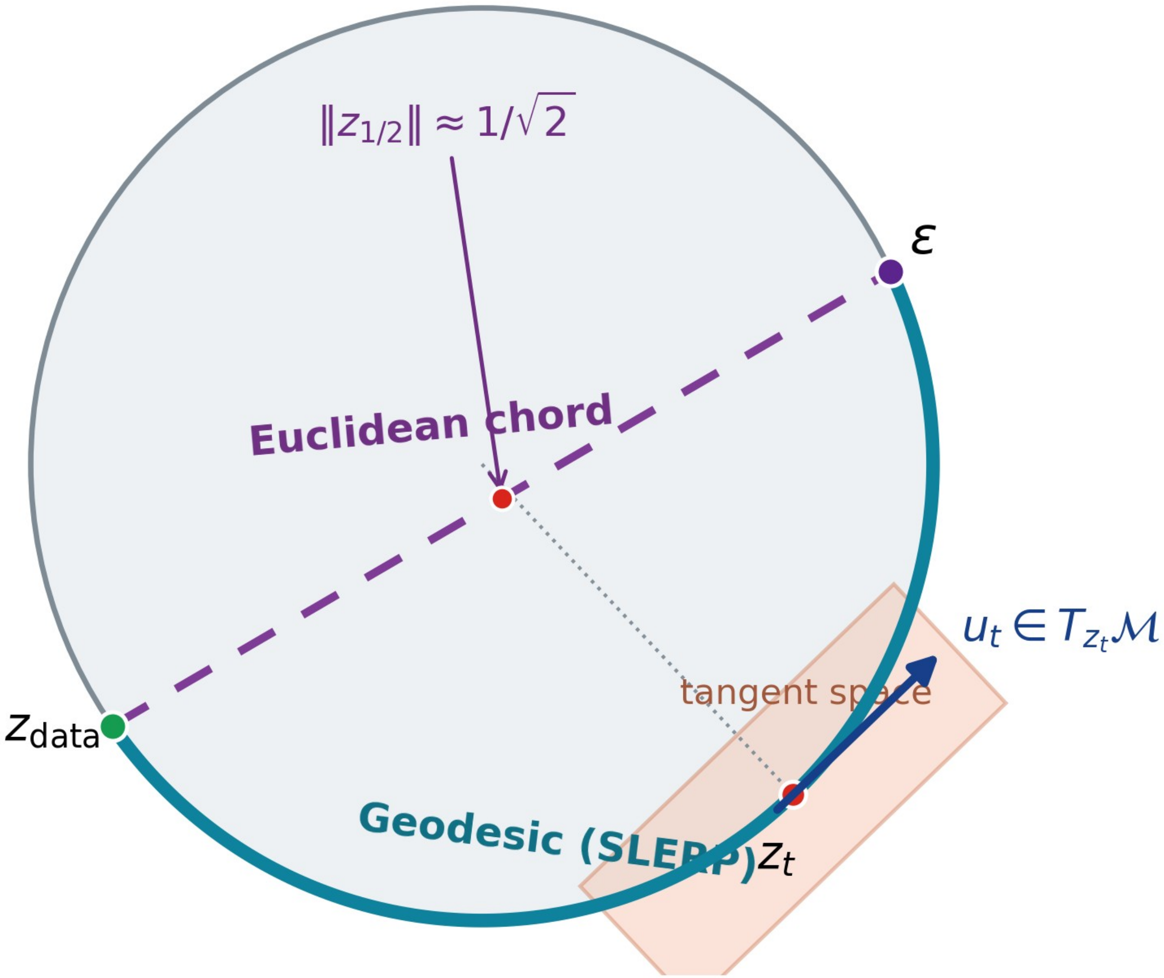
An autoencoder does not have to produce Gaussian latents

Normalization can make token norms nearly constant
(e.g., Transformer + LayerNorm)

The prior can mismatch the latent support or leave it along the transport path

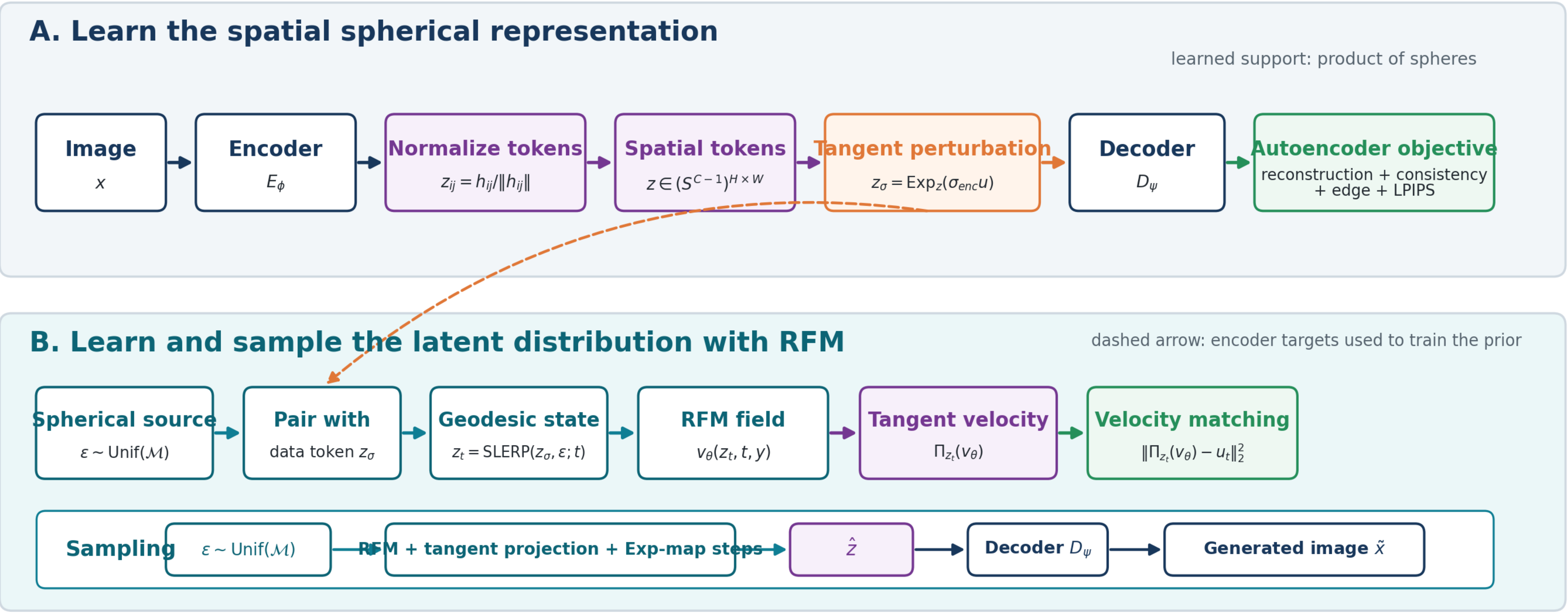
Main contribution
SRUL learns unit-norm spatial tokens and an RFM prior
on the same latent support

CIFAR-10 • PathMNIST • CelebA-64



SRUL pipeline

Learn the spherical representation, learn its distribution, then sample on the same manifold

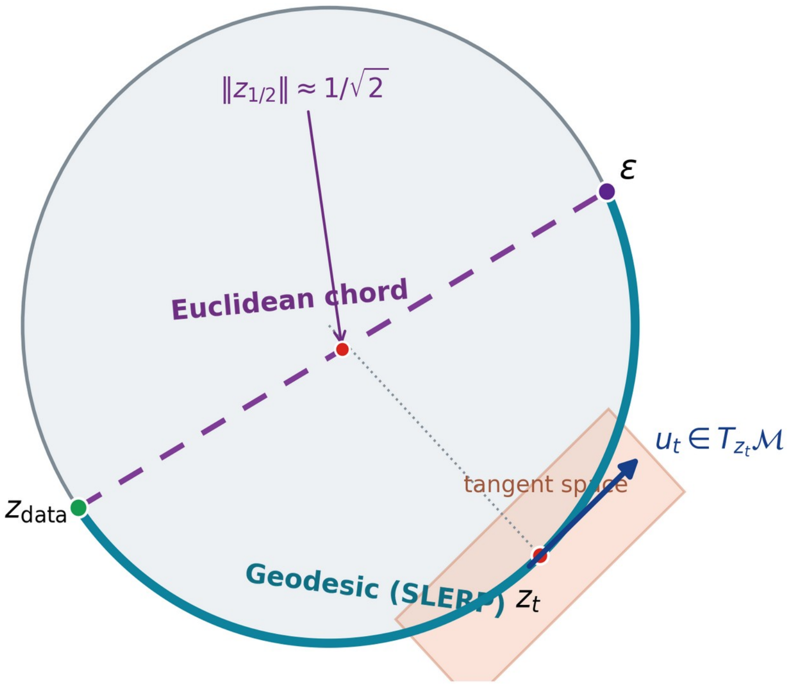


Core geometry: token normalization + tangent perturbation + SLERP + tangent projection + exponential-map integration.

Why the autoencoder and prior must match

Two controlled tests separate path mismatch from support mismatch

Same spherical endpoints, different transport path



1. Path match: same spherical autoencoder

Method	FID	KID	Prec.	Recall	Min norm
Chord	67.44	0.0740	0.840	0.085	0.705
SRUL	**63.84**	**0.0667**	**0.857**	**0.091**	**1.000**

2. Support match: SRUL vs. the same VAE with two priors (CFG = 2)

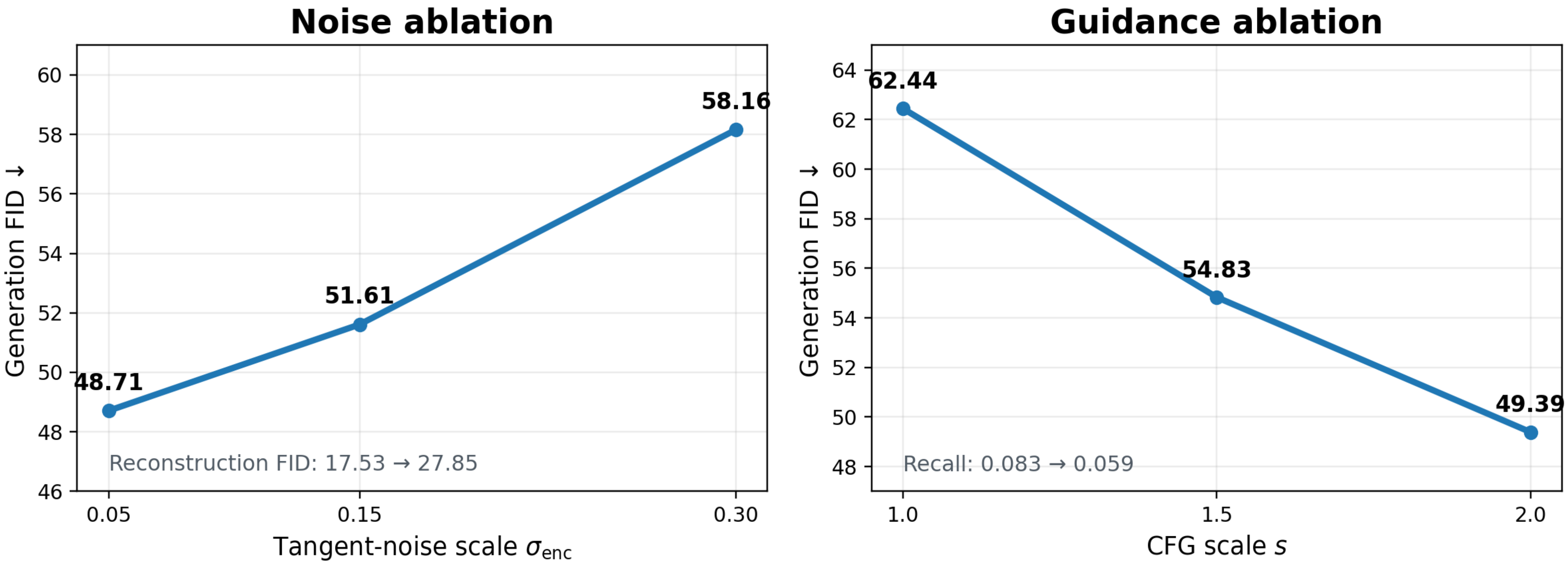
Method	FID	Prec.	Recall
SRUL: spherical AE + RFM	**49.42**	**0.857**	**0.110**
VAE + standard FM	50.70	0.855	0.091
VAE + projected RFM	56.04	0.848	0.063

Same VAE: radius CV = 0.291; mean-radius replacement changes reconstruction FID 16.90 → 22.46.

SRUL learns spherical support before RFM. Projecting a VAE after training removes radius information.

Two controls after the geometry is chosen

Tangent noise changes the latent representation; classifier-free guidance changes sampling

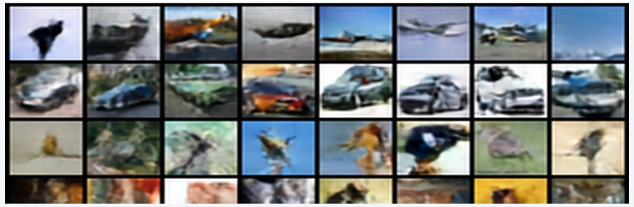


Tangent-noise sweep: σenc = 0.05 gives the best tested reconstruction and generation quality.

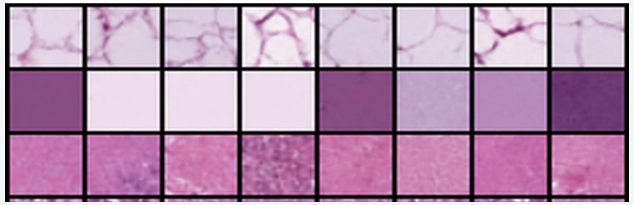
CFG sweep: stronger guidance improves FID and precision, while recall decreases.

SRUL in three image settings

The same method is tested on different image structure and resolution



CIFAR-10



PathMNIST

CelebA-64

Dataset	Evaluation setting	Grid	Recon. FID	Gen. FID	Prec.	Recall
CIFAR-10	Geometry + sampling ablations	4 × 4	17.53	48.71	0.861	0.090
PathMNIST	Histopathology transfer	4 × 4	7.13	29.36	0.829	0.309
CelebA-64	64 × 64 face generation	8 × 8	9.08	41.22	0.657	0.198

CIFAR-10 tests geometry and controls; PathMNIST tests texture transfer; CelebA-64 tests 64 × 64 faces.